EMG Robotic Hand – Hardware Connection Map

How the board, microcontrollers, host and actuators wire together · rev E

BATTERIES
2×9 V in series
±9 V analog

±9 V · J4

EMG FRONT-END PCB
rev E · 100×100 mm · 2-layer
AD623 + MCP6004
· ×330 gain
driven-right-leg · 442 Hz filter

BODY
forearm muscle
3 electrodes
EL+ · EL− · REF

3 electrodes · J1
μV signal

Raspberry Pi Pico
· socketed at J2
ADC0 @ 1 kHz · con-
trol loop · GP0→PWM

+5 V · J10

5 V PSU
claw power

servo lead · J5
GND / +5V / sig

CLAW
1 servo
proportional grip
squeeze ∝ effort

USB · EMG stream
host is observer – loop stays on the Pico

CAMERA
USB webcam

USB video

HOST
Raspberry Pi 5 / Mac
scope · calibrate · log
vision + IK (arm mode)

serial · 1 Mbaud

ESP32
real-time
servo control
200 Hz · fail-safe

6× servos

3-JOINT ARM
+ claw
MG996R servos
230 mm reach

*EMG = **intent** (how hard)*
*camera = **precision** (where)*
fused at the Pi 5

power

6 V PSU
arm power
· ∼10 A

SUBSYSTEM A – built & verified

EMG → claw · rev E board + Pico + host

SUBSYSTEM B – proposed extension

+ camera + 3-joint arm (from the gripper project)

LINE solid = built & verified · dashed = proposed integration

COLOUR EMG signal (μV) · servo / actuation · power · data / USB

Designed & bench-tested in software / CAD – never fabricated, assembled, or powered.
***Battery only** whenever electrodes are on skin: the device never shares ground with the mains.*